

Article

Manufacturing Management Processes Integration Framework

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Abstract

This paper proposes a novel and comprehensive framework for the integration of manufacturing management processes, spanning strategic and operational levels, within and across organizational boundaries. The framework combines a robust set of technologies—such as cyber-physical systems, digital twins, AI, and blockchain—designed to support real-time decision-making, interoperability, and collaboration in Industry 4.0 and 5.0 contexts. Implemented and validated in a Portuguese manufacturing group comprising three interoperating factories, the framework demonstrated its ability to improve agility, coordination, and stakeholder integration through a multi-layered architecture and modular software platform. Quantitative and qualitative feedback from 32 participants confirmed enhanced decision support, operational responsiveness, and external collaboration. While tailored to a specific industrial setting, the results highlight the framework's scalability and adaptability, positioning it as a meaningful contribution toward sustainable, human-centric digital transformation in manufacturing environments.

Keywords: manufacturing management; integration; industry 4.0



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1. Introduction

In recent years, the manufacturing industry has been revolutionized by the convergence of emerging digital technologies, leading to what is widely recognized as the Industry 4.0 paradigm. This transformation is characterized by the integration of cyber-physical systems (CPS), the Internet of Things (IoT), cloud computing, big data analytics, and artificial intelligence (AI) into manufacturing processes, enabling automation, real-time monitoring, and data-driven decision-making [1–5]. These technologies have not only improved operational efficiency but also paved the way for more flexible, responsive, and intelligent manufacturing systems. However, technical progress alone is not sufficient.

Building upon these advancements, the concept of Industry 5.0 has recently emerged as a complementary vision that expands the focus beyond automation and efficiency. As

emphasized by Breque et al. [2], Industry 5.0 advocates a more human-centric, sustainable, and resilient approach to industrial innovation. It reintroduces the human element into the manufacturing equation, promoting collaboration between skilled operators and intelligent systems, while addressing environmental and social challenges. In this dual context, digital transformation must go hand in hand with organizational adaptability, ethical design, and long-term value creation.

Despite these advances, many companies still face serious challenges when integrating their manufacturing management processes. In most industrial environments, decision-making is still fragmented across functional silos—production, logistics, maintenance, and quality management often operate independently with limited coordination. Moreover, the integration between internal processes and suppliers, business partners, and maintenance teams remains underdeveloped, impeding the formation of agile, collaborative ecosystems [6].

Existing research has proposed various technological solutions to address aspects of this integration, including digital twins, AI-based predictive models, and enterprise platforms. However, as Varela et al. [6,7] and Gil-Vilda et al. [8] point out, the literature remains limited in offering comprehensive, validated frameworks that unify strategic and operational management while enabling interoperability across organizational boundaries. Many contributions lack real-world implementation or are restricted to isolated subsystems, with limited scalability or flexibility in diverse industrial contexts.

To address these challenges, this paper proposes a holistic and modular framework for the integration of manufacturing management processes. Designed to align with the principles of both Industry 4.0 and Industry 5.0, the framework supports end-to-end integration—from strategic planning to operational execution—and promotes collaboration among internal departments and external stakeholders. It incorporates a robust technological stack, including digital twins, CPS, AI/ML algorithms, blockchain-based security, and cloud-edge architectures, to support real-time decision-making, interoperability, and data integrity [9].

The proposed framework was developed through a requirements-driven design process and validated in a real-world industrial setting involving a Portuguese manufacturing group with three interconnected factories. A mixed-methods validation strategy, combining quantitative data from stakeholder questionnaires and qualitative feedback from operators and managers, demonstrated improvements in agility, coordination, and cross-functional collaboration.

This study contributes to the field by addressing key limitations identified in the literature, offering a validated and scalable solution for integrated manufacturing management. It further reflects on the organizational, technical, and human factors essential to driving sustainable digital transformation in modern industrial ecosystems.

The remainder of this paper is organized as follows: Section 2 presents a summarized literature review on the key technologies and tools enabling the integration of manufacturing management processes, based on the findings of a systematic literature review; this includes a classification of the studies by technology type, application context, results, and identified limitations. Section 3 introduces the proposed framework, its design rationale, and architectural components. Section 4 details the implementation and validation methodology. Section 5 discusses the key findings and practical implications. Finally, Section 6 concludes with a summary of contributions and suggestions for future research directions.

2. Literature Review

2.1. General Overview

The integration of manufacturing management functions has long been a critical goal for improving efficiency, flexibility, and competitiveness in industrial organizations. Traditionally, efforts in this area focused on aligning internal functions—such as production, maintenance, quality, and logistics—through approaches like just-in-time (JIT) and total productive maintenance (TPM), which sought to reduce waste, optimize resources, and streamline workflows [3]. These approaches aimed to create a more streamlined production flow by reducing waste and optimizing equipment utilization.

Information technology (IT) has played a transformative role in recent years. The introduction of enterprise resource planning (ERP) systems represented a significant leap forward by centralizing data and enabling cross-functional coordination [4]. However, ERP systems often proved rigid and costly, with limited adaptability to real-time decision-making or dynamic production environments. This created a gap between planning and execution layers, especially in complex or distributed industrial settings.

The emergence of Industry 4.0 has transformed the way in which manufacturers approach integration, introducing a set of enabling technologies—including cyber-physical systems (CPS), the Industrial Internet of Things (IIoT), digital twins, cloud computing, and AI/ML algorithms—that allow for decentralized, intelligent, and real-time decision-making [1–5]. These tools facilitate the development of interconnected manufacturing ecosystems, where systems can communicate, adapt, and optimize autonomously based on live data. The impact of these technologies on manufacturing competitiveness and responsiveness has been widely recognized [1,5].

Kagermann et al. [1] provide a retrospective and future outlook on the Industry 4.0 initiative, highlighting its principles, technological pillars (e.g., CPS, IoT), and impact on global manufacturing competitiveness, referring to the ongoing automation of traditional manufacturing and industrial practices using smart technologies like cyber-physical systems, IoT, and cloud computing.

More recently, the Industry 5.0 concept has emerged, driven by the European Commission, to complement the technological focus of Industry 4.0 with a greater emphasis on human-centricity, resilience, and sustainability. As Breque et al. [2] argue, Industry 5.0 prioritizes social and environmental goals, encouraging collaboration between humans and intelligent systems, while promoting inclusive innovation and responsible industrial development.

However, despite this technological evolution, many companies still struggle to fully integrate their manufacturing management processes across both vertical and horizontal dimensions. The need for comprehensive vertical integration (linking strategic levels to operational levels) and horizontal integration (across supply-chain and external actors) has become more urgent in today's volatile and globally distributed manufacturing environments.

Building upon this context, the next section analyzes the most cited technologies, implementation contexts, and integration challenges in the current literature.

2.2. Systematic Literature Review Methodology

To explore these challenges, a systematic literature review (SLR) was conducted using the SCOPUS database, focusing on peer-reviewed journal articles and conference proceedings published between 2014 and 2024. The SLR followed the classification of studies by technology type, application context, results, and limitations.

The inclusion criteria required that studies explicitly addressed the integration of manufacturing management processes and involved the application of enabling technologies

aligned with Industry 4.0 paradigms—such as digital twins, cyber-physical systems (CPS), artificial intelligence, and blockchain.

Keywords were grouped into three categories (see Table 1), ensuring the relevance of results across technological, functional, and management dimensions. Studies were selected based on their contribution to at least one of the following aspects: integration architecture, application context, implementation outcomes, or identified limitations. Priority was given to works that offered conceptual frameworks, reported empirical validation, or highlighted technological innovation applicable to real industrial settings. This selection aimed to provide a representative and diverse overview of the state of the art in integrated manufacturing management.

Table 1. Keyword groups (KWG) considered in the SLR process.

Keyword Group	KWG1	KWG2	KWG3
List of keywords	Manufacturing, Management, Maintenance, System, Process, Cyber-physical, Network, Environment, Factory, Company, Enterprise	Dynamic, Distributed, Parallel, Extended, Integrated, Real-Time, Agile, Intelligent	Artificial Intelligence, IoT, Technology, Augmented Reality, Virtual Reality, Mixed Reality, Digital Twins.

The SLR conducted reflects a steady rise in research focusing on the integration of manufacturing processes, combining different manufacturing management paradigms based on dynamic, distributed or parallel, intelligent and real-time based approaches, along with the use of recent technology and tools in the scope of Industry 4.0, reflecting the need for integrated approaches in modern manufacturing settings (Varela, et al., 2022, 2023) [6,7]. Moreover, the review shows also that such topics have become a focal point for improving sustainability and operational efficiency [8,10].

As illustrated in Figures 1 and 2, the number of publications in this field has steadily increased over the past decade, with MDPI, Elsevier, and IEEE standing out as major contributors. The stacked format of the graph in Figure 2 highlights the contributions of each main editor. Notably, MDPI has experienced significant growth in publications related to manufacturing integration since 2019, reflecting the topic’s multidisciplinary relevance and practical orientation.

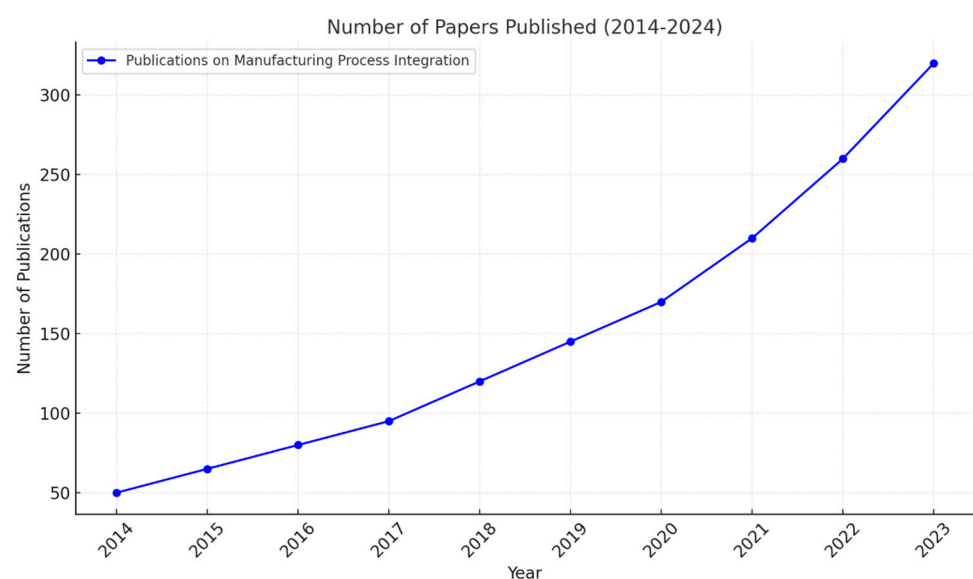


Figure 1. Number of papers published about manufacturing processes integration from 2014 to 2024.

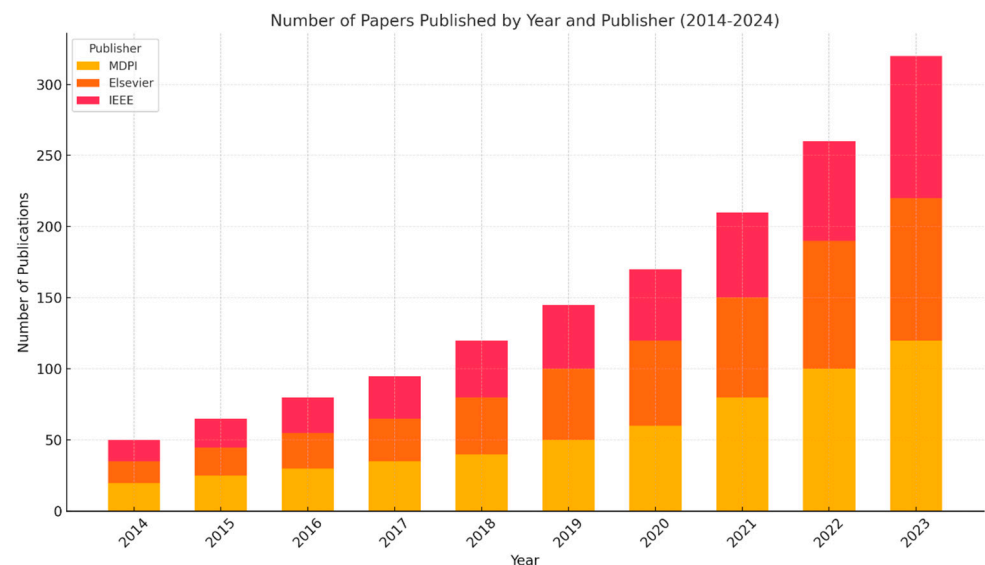


Figure 2. Number of publications per year, categorized by major publishers: MDPI, Elsevier, and IEEE.

This analysis suggests that manufacturing management process integration is a growing field.

Table 2 summarizes the publication trends and key contributions of the main academic publishers—MDPI, Elsevier, and IEEE—in the field of manufacturing management process integration over the last decade.

Table 2. Summary of publisher contributions to the manufacturing management process integration literature (2014–2024).

Publisher	Contribution Trend	Main Characteristics	Key Insight
MDPI	Rapid increase since 2019	Strong focus on multidisciplinary and applied research; publishes extensively on Industry 4.0 topics	Rapid growth reflects the success of its open-access model and its appeal to fast-evolving research areas
Elsevier	Steady over the decade	Established reputation; large portfolio in engineering, manufacturing, and management	Remains a cornerstone for both foundational and applied research in the field
IEEE	Steady but smaller increase	Emphasis on technological and engineering innovations; focus on core enabling technologies like CPS and IoT	Plays a key role in technology-centric contributions, though with fewer total publications

2.3. Analysis and Discussion of Key Contributions from the Literature

Despite the positive trend observed in recent years, the literature still reveals important gaps. The conducted SLR confirms a substantial increase in research over the past decade focused on the integration of manufacturing management functions, particularly within the context of Industry 4.0. Key areas of focus include collaborative manufacturing, cross-functional integration in product and process development, and the incorporation of total quality management (TQM) with advanced digital technologies.

Recent trends highlight the growing adoption of emerging digital technologies and tools, not only in traditional manufacturing settings but also in extended, virtual, and distributed environments. These include cyber-physical systems (CPS), the Internet of Things (IoT), integration platforms, digital twins, artificial intelligence and machine learning (AI/ML), and blockchain, used to support real-time decision-making and enhance process efficiency and collaboration.

Publications have increased notably in recent years, reflecting the adoption of new management paradigms—such as dynamic, distributed or parallel, intelligent, and real-time-based approaches—alongside concerns about collaboration and sustainability [11].

Table 3 provides a synthesis of the most relevant contributions identified in the SLR, classified according to the enabling technologies applied, the industrial or strategic application contexts, the key outcomes reported, and the limitations observed. This classification complements the bibliometric analysis by contextualizing the studies in terms of practical scope and relevance to integrated manufacturing management.

Table 3. Synthesis of main contributions on integrated management processes.

Authors	Main Contributions	Source (Journal/Editor)
Roeck, D. (2020) [12]	Refers to the Foundation of Distributed Ledger Technology for Supply-Chain Management	HICSS, ScholarSpace
Koulamas, C. and Lazarescu, M.T. (2020) [13]	Study real-time sensor networks and systems for the industrial IoT.	Journal of Manufacturing Science and Engineering, ASME
Zhou et al. (2021) [14]	Study intelligent small object detection for digital twin in smart manufacturing with industrial cyber-physical systems.	IEEE Transactions on Industrial Informatics
Simion et al. (2021) [15]	Research on the Use of Integrated Management Systems	Advances in Science and Technology
Kagermann, H., Wahlster, W. and Helbig, J. (2022) [1]	Discusses the key technologies and concepts driving Industry 4.0 and their impact on manufacturing.	Sci, MDPI
Singh et al. (2022) [16]	Focus on the applications of digital twin across industries, based on a literature review.	Applied Sciences, MDPI
Flores-García et al. (2023) [17]	A study on enabling digital servitization based on the Industrial Internet of Things (IIoT) in smart production logistics.	International Journal of Production Research, Taylor & Francis
Wang, H., Yan, Q. and Wang, J. (2023) [18]	Refer to blockchain-secured multi-factory production with collaborative maintenance using Q learning-based optimization approach.	International Journal of Production Research, Taylor & Francis
Grievies (2024) [19]	Intelligent digital twins and the development and management of complex systems	Digital Twin, Taylor & Francis
Li et al. (2024) [20]	Refer to a digital twin system for Task-Replanning and Human-Robot control of robot manipulation	Advanced Engineering Informatics, Elsevier.
Zheng et al. (2024) [21]	Distributed Energy Management of Multi-Entity Integrated Electricity and Heat Systems: A Review of Architectures, Optimization Algorithms, and Prospects	IEEE Transactions on Smart Grid
Ghughe, Akarte and Raut, (2024) [22]	Decision-making frameworks in additive manufacturing management: mapping present landscape and establishing future research avenues	Benchmarking: An International Journal
Khan et al. (2024) [23]	Measuring economic resilience of manufacturing organization leveraging integrated data envelopment analysis (DEA)-machine learning approach	International Journal of Management Science and Engineering Management
Kim et al. (2024) [24]	Techno-economic analysis for design and management of international green hydrogen supply chain under uncertainty: An integrated temporal planning approach	Energy Conversion and Management
Hammad, M. et al. (2023) [25]	Propose a framework using fuzzy DEMATEL to support smart manufacturing implementation in SMEs.	Processes, MDPI
Vacchi, M. et al. (2024) [26]	Propose a strategic framework for assessing technological sustainability to support the transition from Industry 4.0 to 5.0.	Sustainability, MDPI
Martín-Gómez, A.M. et al. (2024) [27]	Propose a framework integrating Industry 4.0 technologies with Industry 5.0 values for sustainable manufacturing.	Sustainability, MDPI
Wang, Y. et al. (2024) [28]	Present an intelligent management system for small-scale industries, integrating IoT and optimization algorithms.	Electronics, MDPI
Raddi-Mira, M. et al. (2024) [29]	Assess maturity models for smart factories in the context of Industry 5.0, identifying evaluation gaps and priorities.	Sustainability, MDPI
Lyngdorf, A. et al. (2024) [30]	Review and structure smart maintenance within Industry 5.0 using a PRISMA-based systematic literature review.	Education Sciences, MDPI
Bastos, F. et al. (2024) [31]	Present a knowledge-based model to evaluate a framework of the best practices to drive the digital transition in industry's	Future Internet, MDPI
Maria, H. et al. (2025) [32]	Integrated approach for evaluating maturity models in smart factories, aligning with Industry 5.0 values.	Technological Forecasting and Social Change, Elsevier
Dehghan, S. et al. (2025) [33]	Provide a bibliometric analysis of the integration of additive manufacturing into Industry 4.0 and 5.0 paradigms.	Machines, MDPI
Md Tariqul Islam et al. (2025) [34]	Identify key challenges and enabling technologies for Smart Manufacturing transition through a systematic literature review.	Machines, MDPI

The selected works focus on ‘Integration’, alongside other key management paradigms such as ‘Dynamic’, ‘Distributed’, and ‘Real-time based’ approaches within manufacturing systems. These studies highlight the use of enabling technologies and tools—including digital twins (DTs), cyber-physical systems (CPS), blockchain, artificial intelligence/machine learning (AI/ML), and integration platforms. These technologies have been applied not only in traditional manufacturing environments but also in extended, distributed, and virtual manufacturing contexts. The latter, in particular, poses greater demands in terms of technological sophistication and interoperability, requiring advanced solutions aligned with Industry 4.0 principles. The role is critical for supporting informed and collaborative decision-making among diverse stakeholders, optimizing production processes, ensuring real-time responsiveness, and fostering cooperation within complex global manufacturing ecosystems.

A substantial number of studies remain limited to conceptual frameworks, simulations, or technology-specific applications that lack validation in real-world industrial environments (e.g., [14,18,35]). Recurring challenges include poor interoperability among heterogeneous systems, concerns over data security in open networks, and limited integration between strategic planning and operational execution on the shop floor. Additionally, few studies effectively address the collaborative and cross-organizational nature of decision-making within extended manufacturing ecosystems.

As emphasized by Simon et al. [15] and Khan et al. [23], effective integration demands more than just technical interoperability—it also requires process alignment, cultural readiness, and organizational maturity. These gaps reinforce the urgent need for validated and scalable frameworks that combine technological innovation with human-centric design and real-world applicability—which is precisely the motivation behind the framework proposed in this study.

The literature review reveals a strong emphasis on the technological dimension of manufacturing integration; however, several fundamental aspects remain underdeveloped. Although vertical and horizontal integration are frequently mentioned as priorities, few frameworks simultaneously address both. Moreover, there is a clear gap between theoretical contributions and real-world applications, with many proposals lacking empirical validation in industrial contexts. Interoperability and cybersecurity persist as critical technical challenges, especially in multi-factory or multi-stakeholder environments. Furthermore, the emerging values of Industry 5.0—such as human-centricity and sustainability—are not yet fully embedded into most integration models, revealing the need for more holistic and inclusive approaches.

Thus, the main technologies that have been used in the last decade and were reported by the authors (Table 3) provide various solutions to challenges faced by modern, distributed manufacturing systems. In this context, digital twins (DTs) and cyber-physical systems (CPS) are mentioned among the current technologies that are particularly significant for enabling the real-time monitoring, simulation, and optimization of production processes and integration. Digital twins create virtual representations of physical assets, enabling manufacturers to predict, model, and optimize performance in dynamic environments. Cyber-physical systems facilitate the seamless communication and synchronization of physical systems with digital counterparts, driving operational improvements across production sites.

Blockchain technology adds a critical layer of security, ensuring transparency and trust in decentralized manufacturing environments. It helps to secure data exchanges and enables collaborative efforts without concerns over data integrity. Moreover, AI/machine learning (AI/ML) tools are key technologies for predictive analytics, anomaly detection,

and real-time decision-making, offering an edge in optimizing production efficiency and managing resources dynamically.

Integration platforms, such as cloud-based solutions and Industrial IoT (IIoT) platforms, play a fundamental role in connecting all the mentioned technologies. These platforms enable real-time data exchange, smooth integration, and data-driven decision-making across distributed manufacturing systems, ensuring operational continuity and improved scalability.

While significant progress has been made, important gaps remain—particularly in interoperability, scalability, and the integration of human-centered values. To better understand these gaps, the next section classifies the enabling technologies and their limitations. The full enablement of the transition to Industry 4.0 requires the tackling of challenges, such as data security, very carefully, along with interoperability issues between disparate systems, and fostering a culture of collaboration across all organizational levels. Additionally, research should explore how to leverage advanced analytics and artificial intelligence to optimize decision-making processes, namely in the manufacturing management domain.

2.4. Classification of Enabling Technologies, Applications and Integration Gaps

This section classifies the technologies used, the contexts where they are applied, and the main problems found in the literature.

Beyond the bibliometric trends and analysis made in the previous section, the reviewed studies in the literature were analyzed to provide an analytical classification of the enabling technologies, their application contexts, and the limitations observed in their implementation across manufacturing management scenarios.

2.4.1. Technology Classification

The reviewed studies were categorized according to the primary enabling technologies addressed. The most frequently discussed technologies include:

- Cyber-physical systems (CPS) and digital twins (DTs) support real-time monitoring, system simulation, and adaptive control within manufacturing processes [13–19,36];
- Internet of Things (IoT) and cloud/edge computing, providing decentralized data acquisition and connectivity for distributed decision-making [5,37];
- Artificial intelligence and machine learning (AI/ML), used for predictive analytics, demand forecasting, anomaly detection, and quality control [5,23,24,36,38];
- Blockchain technology, employed to ensure secure, immutable, and transparent information exchange across distributed networks [18,35–38];
- Integration platforms, which serve as middleware to orchestrate workflows between legacy systems and modern digital infrastructures [12,16,39,40].

2.4.2. Implementation Contexts

The majority of the implementations occurred within smart factories and extended enterprises, typically operating across multiple physical locations or interconnected digital ecosystems. Several studies targeted discrete manufacturing sectors, such as automotive or electronics, while others focused on virtual manufacturing networks and collaborative supply chains [20,41].

Integration initiatives addressed both vertical integration (linking strategic, tactical, and operational levels within organizations) and horizontal integration (across external stakeholders such as suppliers, partners, and customers), revealing a clear trend toward holistic and interoperable manufacturing ecosystems [38].

2.4.3. Identified Limitations

Despite notable advances, the literature reveals several recurring limitations that hinder the full deployment and scalability of integration frameworks:

- Interoperability issues persist due to the heterogeneity of technologies, standards, and data formats used across systems and organizations [1,39];
- Scalability challenges, especially in adapting proposed solutions to diverse industrial scales, operational models, or legacy infrastructures [35,42–44];
- Cybersecurity concerns, particularly when handling sensitive or proprietary data across open, distributed ecosystems [5,14,36,45];
- Human-related barriers, including workforce resistance to digital transformation and insufficient technical skills to manage complex digital systems [2,20];
- Limited validation, with many studies relying on simulation environments, prototypes, or expert opinion, rather than real-world deployments with measurable operational outcomes [11,14].

Building upon the identified limitations, it becomes essential to position our proposed solution relative to existing industry frameworks.

These limitations highlight the ongoing disconnect between conceptual frameworks and their effective implementation in complex industrial environments.

2.4.4. Comparative Positioning with Reference Frameworks

To clarify the novel aspects of our proposed framework, a comparative analysis was conducted against widely recognized industrial reference models, such as RAMI 4.0, ISA-95, and the Industrial Internet Reference Architecture (IIRA). These models provide robust conceptual structures but often lack guidance for implementation in complex, real-time, and scalable industrial scenarios [46].

As summarized in Table 4, our framework extends these models by introducing a more implementation-driven structure with clear support for edge processing, AI-based services, human–machine interaction, and compliance with Industry 5.0 values such as resilience and user-centricity.

Table 4. Comparative overview of selected frameworks and the proposed integration approach.

Framework	Focus	Real-Time	Edge/AI Integration	Human-Centric	Implementation Level
ISA-95	Enterprise Control Levels	X	X	X	Conceptual
RAMI 4.0	Asset Lifecycle and Layers	●	●	X	Reference
IIRA	Domains and Functional View	●	●	X	Reference
Proposed	Integrated Execution Model	✓	✓	✓	Practical, Deployable

2.5. Implications

Recent studies have further advanced the practical application of Industry 4.0 and 5.0 technologies through intelligent systems for quality control, simulation, and adaptive robotics. For example, computer vision and deep learning techniques have been applied to real-time object detection and manufacturing inspection systems using YOLO architectures and custom datasets for industrial applications. Simulation environments for optimizing industrial processes, such as robotic painting or trajectory planning, have demonstrated the integration of virtual models with production constraints. Furthermore, robotic systems equipped with advanced control and feedback mechanisms are being explored for autonomous operations in dynamic shop floor environments, reinforcing the importance of interoperability and decision-making in cyber-physical systems. These works complement the proposed framework by providing evidence of successful implementations of modular, AI-enabled industrial systems [47].

Recent studies have emphasized the importance of integrating digital maturity, sustainability, and human-centric approaches in industrial systems. These contributions support the need for interoperable, intelligent, and policy-aware frameworks aligned with Industry 4.0 and 5.0 principles, reinforcing the motivations behind the approach adopted in this work [48].

This analytical classification highlights a critical gap: although technological advancements are abundant, few solutions address the intersection of technical innovation, organizational maturity, and human-centered design. This gap reinforces the need for integrated, validated, and flexible frameworks that align technological capabilities with strategic objectives, facilitate interoperability, and incorporate user-centric principles.

The framework proposed in this paper aims to address these gaps, offering a modular and scalable approach that supports real-time decision-making, cross-organizational collaboration, and secure, interoperable communication among stakeholders, fully aligned with both Industry 4.0 and Industry 5.0 paradigms.

The preceding analysis reveals several limitations in current frameworks and implementations. These gaps directly informed the conceptualization of our proposed framework, which is detailed in the following section.

2.6. Implications

The enabling technologies and integration principles discussed throughout this review have been predominantly applied in sectors such as automotive, electronics, and aerospace. However, recent advancements demonstrate that sectors traditionally less associated with digital transformation, such as the textile industry, are also embracing Industry 4.0 and 5.0 paradigms [49–53].

In particular, several studies have shown that computer vision, image processing, and deep learning are being effectively employed for yarn quality monitoring, defect detection, and production traceability. For example, Pereira et al. [49–53] implemented convolutional neural networks to evaluate yarn characteristics in real-time, enabling more accurate and automated quality control. In a related work, image-based linear mass estimation was demonstrated as a viable method for replacing manual inspection processes.

These examples highlight the increasing maturity of the textile industry in adopting intelligent systems and show how integration frameworks must support domain-specific challenges such as continuous flow manufacturing, visual quality standards, and high production throughput. As such, the framework proposed in this study is well-suited for adaptation to textile manufacturing contexts, where real-time sensing, AI-driven analytics, and edge–cloud integration are becoming essential.

Moreover, these studies reinforce the broader applicability of the integration approach presented, offering a path forward for digitally transforming not only discrete manufacturing systems but also process-oriented and hybrid industries.

Building upon the gaps and opportunities identified in the literature, the next section introduces a modular and integrative framework aimed at addressing these challenges. The framework is designed to align with the operational, technological, and human-centered dimensions emphasized across Industry 4.0 and Industry 5.0 paradigms [54].

3. Framework for Manufacturing Management Integration

The proposed framework adopts a layered architecture to support the integration of manufacturing management processes, ranging from real-time data acquisition to strategic decision-making. This conceptual structure is aligned with industrial standards such as RAMI 4.0 and ISA-95, ensuring modularity, scalability, interoperability, and security [55–70].

Figure 3 illustrates the conceptual layered architecture, structured into four distinct layers:

- **Device Layer:** Includes physical assets (sensors, RFID readers, machines) responsible for data generation at the shop floor level;
- **Edge/Integration Layer:** Manages local data processing, filtering, and protocol conversion (e.g., OPC-UA, MQTT), ensuring semantic consistency and low-latency responsiveness;
- **Platform Layer:** Provides core functionalities such as data storage (data lake), analytics, and microservices orchestration, deployed via containerized environments;
- **Application Layer:** Delivers dashboards, alerts, and decision-support tools, ensuring secure and human-centric interactions.

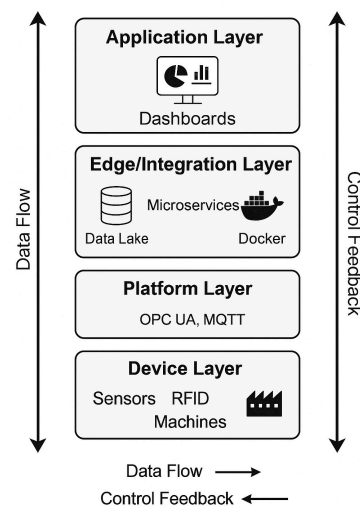


Figure 3. Conceptual layered architecture of the proposed integration framework, structured into Device, Edge/Integration, Platform, and Application layers.

To demonstrate the framework’s applicability in a real industrial environment, Figure 4 presents its implementation using Microsoft Azure services. This practical deployment includes user authentication, SQL database and data lake integration, machine learning tools, digital twins, and multiplatform data visualization.

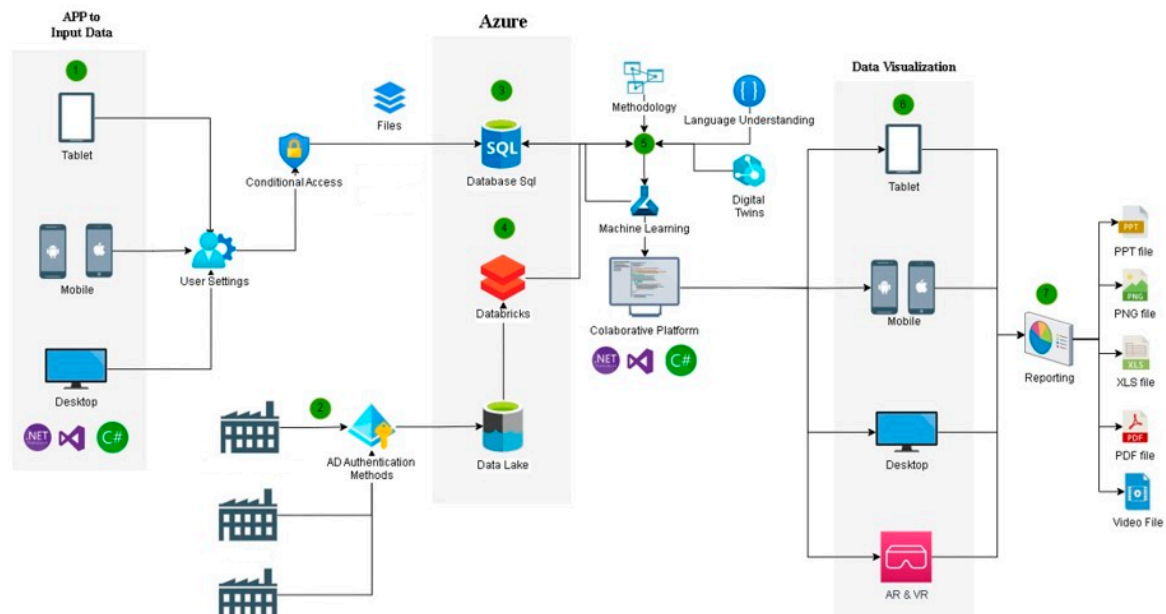


Figure 4. Practical implementation of the proposed framework using Microsoft Azure services. It illustrates user input from multiple devices, secure authentication, centralized data storage and processing, advanced analytics tools, and data visualization across multiple formats.

This dual perspective—both conceptual and practical—highlights both the theoretical rigor and the technological feasibility of the proposed integration framework.

3.1. Framework Requirements and Design Rationale

Building upon the integration challenges identified in the literature, the proposed layered architecture addresses key gaps related to modularity, real-time responsiveness, and interoperability.

The design of the framework is guided by a set of core requirements derived from industry needs and stakeholder input:

- **Integration Complexity:** Coordination across internal departments and external stakeholders requires a unified data and decision-making backbone;
- **Real-Time Coordination:** Operational decisions must react to events in real time, therefore, the system must process and act upon data with minimal delay;
- **Interoperability:** Seamless interaction between legacy systems (e.g., ERP, MES) and modern digital tools is essential;
- **Data Security and Privacy:** As data cross organizational boundaries, secure communication (e.g., encrypted APIs, blockchain integrity) is required;
- **Scalability and Adaptability:** The architecture must be modular and scalable across different sites and adaptable to new industrial contexts.

These requirements are fulfilled through the layered architecture shown in Figure 3 and realized technically in the Azure-based implementation shown in Figure 4.

The practical implementation includes the following technological enablers:

- **IoT and Edge Components:** For real-time sensing and local decision-making;
- **Integration Middleware:** To bridge legacy systems with modern services;
- **Machine learning and Digital Twins:** For predictive analytics and intelligent monitoring;
- **Blockchain-based Security Mechanisms:** To ensure traceability and trust in data transactions across multiple stakeholders;
- **User Interfaces and Dashboards:** Designed for intuitive decision support, promoting human-in-the-loop collaboration in line with Industry 5.0 principles.

The framework was designed not only to address the typical demands of Industry 4.0, such as automation, real-time control, and data-driven optimization, but also to align with the emerging principles of Industry 5.0. These include sustainability, resilience, and human-centric system design. As such, the architecture incorporates adaptive interfaces, energy-efficient components, and secure multi-stakeholder integration to ensure that technological solutions remain both agile and socially responsible.

Together, these components enable a modular, interoperable, and secure environment for managing manufacturing processes at multiple organizational levels and across supply-chain partners.

To address these requirements in practice, the next section details the technological components selected for implementation.

3.2. Technological Components and Tools

The platform was implemented using proven tools and technologies that are scalable, secure, and aligned with Industry 4.0 and 5.0 needs.

The implementation leverages widely adopted and open technologies, selected for their compatibility, scalability, and reliability:

- **IoT Integration:** Utilized MQTT protocol and Node-RED for data orchestration, along with microcontroller platforms (e.g., Arduino, ESP32) for sensor-based input;

- *Backend Logic*: Developed using Python (Version 3.11) and Node.js (Version 20.x) to ensure flexibility and easy API integration;
- *Data Storage and Streaming*: PostgreSQL and Apache Kafka support high-volume data management and real-time processing;
- *AI/ML Analytics*: Implemented using Scikit-learn and TensorFlow for predictive modelling and adaptive learning;
- *Dashboards and Interfaces*: ReactJS and Grafana were used to create rich visualization environments for decision support;
- *Cloud Infrastructure*: Hosted on Microsoft Azure to ensure scalability and secure access across factories;
- *Security and Privacy*: JWT-based authentication and a blockchain pilot layer provide traceability and protection of sensitive data.

3.2.1. Software

The developed software platform integrates advanced technologies from Industry 4.0, such as IoT, AI, automation, and big data while adapting them to the values and requirements of Industry 5.0. Industry 4.0 focuses on automation and digitalization, while Industry 5.0 prioritizes human-centric, resilient, and sustainable practices. The software developed combines both perspectives. The focus shifts from purely efficiency-driven automation to intelligent systems that are sustainable, resilient, and human-aware. This reorientation was a core design principle throughout the platform's development.

The software was developed through a structured process, from requirements gathering to iterative implementation and validation, ensuring that it remains both robust and adaptable. It offers functionalities that go beyond automation, actively supporting decision-making, collaboration, and long-term sustainability across industrial contexts:

1. **Integration with IoT:**
 - For connecting devices, sensors and systems, creating a data network that operates in real time;
 - The software facilitates communication between machines (M2M) and between machines and human systems, enabling a seamless and transparent workflow;
 - Application example: Remote monitoring of machines, real-time production control and predictive maintenance.
2. **Intelligent Automation:**
 - The developed platform enables to automate repetitive and complex processes, ensuring precision, speed and error reduction;
 - Automation is supported by advanced algorithms that adjust processes based on real-time and historical data;
 - Application example: optimization of assembly lines and integrated inventory management.
3. **Artificial Intelligence and Machine Learning:**
 - The platform implements AI for predictive analysis, pattern identification and resource optimization;
 - The underlying platform's software uses machine learning to continually improve processes, learning from historical data and adapting to changes in the operational environment;
 - Application example: equipment failure prediction and dynamic production planning.

4. Integrated Management Systems:
 - The platform enables to connect all areas of a company—production, logistics, sales and maintenance—into a single integrated ecosystem;
 - This guarantees full visibility of processes and allows for informed and agile decisions;
 - Application example: interactive dashboards with KPIs (key performance indicators) updated in real time.
5. Advanced Data Analysis and Big Data:
 - The platform's software collects and analyzes large volumes of data, transforming it into actionable information;
 - It uses data visualization techniques to present insights in a clear and intuitive way, facilitating strategic decision-making;
 - Application example: energy efficiency analysis and identification of bottlenecks in the production process.
6. Connectivity with Industry 5.0:
 - The developed platform places humans at the center of the operation, promoting collaboration between machines and operators;
 - It prioritizes personalization, sustainability and an intuitive and fluid user experience;
 - Application example: human-machine interface (HMI) systems with virtual assistants for technical and operational support.

The developed integrated platform is an innovative prototype, designed to meet the needs of Industry 4.0 and Industry 5.0, where the former focuses on automation and digitalization and the latter emphasizes human-centric, resilient and sustainable approaches, aligned with the most modern Internet of Things (IoT) technologies and fully integrated with our advanced software systems. These prototypes represent the perfect convergence between hardware and software, providing solutions that not only optimize industrial processes, but also enable a higher level of connectivity, intelligence and efficiency. In this area, the developed integrated platform enables product design linked to the next area of automation and underlying devices, which is where the technical components involving automation and electronics are interoperated.

3.2.2. Prototypes

The prototypes are aimed at Industry 4.0 and Industry 5.0, where the former focuses on automation and digitalization and the latter emphasizes human-centric, resilient and sustainable approaches.

The developed platform is designed to act as part of a smart, interconnected ecosystem, including:

- Real-Time Monitoring and Control: Devices equipped with advanced IoT sensors to track operational conditions, production and performance in real time;
- Automation and M2M Communication: Prototypes that enable direct communication between machines, optimizing operations and reducing time-consuming human intervention;
- Personalization and Sustainability: Equipment designed to customize operations according to specific needs and promote energy and environmental efficiency.

The main features of the software underlying our proposed integrated platform are characterized by the following main aspects:

1. **Modularity and Scalability:**
The solutions can be customized to meet each client's specific needs, with the possibility of expansion as operations grow;
2. **Compliance with Global Standards:**
The developed platform is aligned with the highest industrial standards, ensuring compatibility with equipment and systems from different suppliers;
3. **Security and Robustness:**
The platform implements advanced cybersecurity protocols to protect critical data and operations from attacks or failures based on blockchain technology;
4. **Interoperability:**
The platform integrates different technology, including legacy systems and new technologies, ensuring that everything works harmoniously and in real time.

3.2.3. Impacts of the Software on the Industrial Environment

Using this this integrated platform, the company can thus:

- **Increase Productivity:** Automating tasks and optimizing processes, reducing waste and downtime;
- **Reduce Operating Costs:** By improving energy efficiency, predicting failures and optimizing resource use;
- **Improve Product Quality:** With greater control over processes and real-time adjustments;
- **Promote Sustainability:** With analysis and optimization of resources, promoting a greener and more economical operation;
- **Increase Competitiveness and interoperation:** By adopting cutting-edge technologies that place the company at the forefront of the sector, enabling interoperation between different technology and further collaboration among the company's interacting factories and other external stakeholders, including suppliers, business partners and clients;
- These impacts demonstrate how traditional Industry 4.0 technologies can be reconfigured to serve broader objectives beyond productivity, including environmental efficiency, system robustness, and collaborative human-machine interaction. This reinforces the alignment of the proposed platform with the paradigm of Industry 5.0, where technology plays a supporting role in more adaptive, inclusive, and sustainable manufacturing ecosystems.

3.3. Platform Development Methodology and Lifecycle

This section details the methodology followed in the development of the proposed software platform. It combines user-centered design practices, agile engineering, and validation in a real industrial environment. The lifecycle encompassed structured requirement gathering, modular system design, iterative implementation, testing, and continuous stakeholder engagement. This approach ensures scientific rigor, replicability, and alignment with industrial standards.

Figure 5 visually summarizes the development lifecycle adopted in this project. Each phase is aligned with engineering best practices and was iteratively validated in close collaboration with industry stakeholders. This structure helped to ensure that the resulting platform is robust, modular, and adaptable to real operational contexts.

The software platform underlying the proposed integration framework was developed following a structured and iterative software engineering approach to ensure both technical robustness and user-centered adaptability. The process included the following key stages:

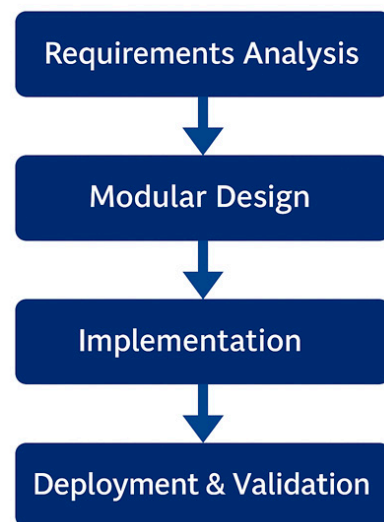


Figure 5. Software development lifecycle of the proposed platform.

- **Requirements Analysis:** Functional and non-functional requirements were gathered through direct interaction with company stakeholders (e.g., production managers, IT staff, and operators). These included real-time monitoring capabilities, user-friendly dashboards, modular system architecture, integration with legacy ERP/MES systems, and security compliance;
- **Modular System Design:** The architecture was modelled using a microservices-based approach. Each functional component (e.g., IoT connectivity, AI analytics, dashboard generation) was designed as an independent module to enhance maintainability and scalability. The system design was documented using UML diagrams and process workflows;
- **Implementation:** The platform was implemented using open-source technologies and widely adopted industrial platforms. Python and Node.js were used for core logic and API handling, while MQTT protocols handled IoT communication. For data handling and analytics, PostgreSQL and Apache Kafka were used. Cloud infrastructure was deployed using Microsoft Azure, supporting elastic scalability;
- **Testing and Integration:** A test-driven development (TDD) approach was adopted. Unit, integration, and system-level tests were performed to validate the functionality and robustness of each module. System simulations using historical and synthetic datasets were conducted to assess performance under various operational loads;
- **Deployment and Maintenance:** The platform was deployed in a real industrial environment—a Portuguese manufacturing group comprising three factories. Post-deployment support included performance monitoring, user feedback integration, and regular patching based on stakeholder suggestions.

To evaluate the effectiveness of the platform after deployment, a set of key performance indicators (KPIs) were monitored in collaboration with the industrial partner. These included:

- Reduction in machine downtime (measured through MTTR—Mean Time to Repair);
- Increase in production throughput per shift;
- Improved detection and response time to operational anomalies;
- Overall system uptime and reduction in integration-related errors.

Although exact values cannot be shared due to confidentiality, the trends showed better coordination, quicker decisions, and greater visibility across the production units.

The validation results suggest that the proposed integration framework not only supports technological scalability and modularity but also leads to measurable operational improvements. Future work will involve a longitudinal analysis of these KPIs and benchmarking against baseline scenarios to further confirm the platform's industrial impact.

This lifecycle ensured that the resulting solution aligned with both theoretical principles and operational needs, while remaining scalable, secure, and adaptable to diverse manufacturing contexts. Furthermore, the use of industry-standard technologies and cloud-native design enables rapid future extension to other production sites or domains.

As a final step in the platform's development lifecycle, a structured questionnaire was used to gather feedback from 16 participants who were directly involved in its implementation. These included IT staff, production managers, and operational users from the partner company.

The questionnaire combined Likert-scale items and open-ended questions to evaluate dimensions such as usability, integration, responsiveness, and strategic relevance. While confidentiality constraints prevent the disclosure of detailed results, aggregated feedback revealed high user satisfaction and positive perceptions of the platform's impact on production coordination and information flow.

This practical validation, supported by qualitative and quantitative insights, contributed to confirming the system's applicability and alignment with user expectations.

Observed Implementation Limitations and Practical Implications

During the industrial deployment of the proposed platform, several practical limitations were encountered. First, the integration with existing legacy systems required customized middleware configurations due to incompatibility with modern APIs and communication protocols. Second, certain operational data necessary for full-scale validation were subject to confidentiality constraints, limiting real-time access and detailed performance benchmarking. Additionally, the adoption process faced organizational barriers, including user adaptation challenges and the need for extended training. From a strategic perspective, variability in infrastructure maturity across the three production sites introduced inconsistencies in deployment timelines and scalability.

These limitations influenced the pace and scope of implementation. For instance, some advanced features—such as dynamic resource optimization—needed to be phased in gradually. Nonetheless, these challenges were partially mitigated through iterative development, stakeholder engagement, and fallback mechanisms for data acquisition. Future work will explore more standardized integration layers, broader organizational alignment, and a longitudinal evaluation of impacts over time.

Following this development lifecycle, it was necessary to validate the platform's effectiveness in a real industrial setting. After implementing the platform, the next step was to evaluate its performance in a real industrial setting, based on feedback from stakeholders.

The architectural and methodological aspects described above were implemented in a real-world industrial context. To assess the framework's practical effectiveness, a validation strategy was devised, and applied, and is presented in the following section.

4. Framework Implementation Validation

The validation stage presented in this section is directly derived from the implementation of the proposed framework described in Section 3. The layered architecture and integration strategies previously defined were translated into a functional platform, whose performance, usability, and user acceptance were assessed through a structured questionnaire. This validation process was designed not as an isolated step, but as a nat-

ural continuation of the conceptual and technological foundation laid earlier, ensuring coherence between the design and its practical impact.

To validate the proposed framework, a structured questionnaire was developed and administered to 32 stakeholders involved in the implementation, including engineers, IT specialists, supervisors, and system integrators. The goal was to assess their perceptions of the platform's effectiveness across key performance dimensions.

The questionnaire consisted of five evaluation categories:

- Interoperability;
- Usability;
- Responsiveness;
- Data integration;
- Alignment with Industry 4.0/5.0 principles.

Each item was rated on a five-point Likert scale (from 1—strongly disagree to 5—strongly agree). The average scores per category are summarized in Figure 5. These results offer valuable insights into the platform's practical utility and perceived performance across relevant dimensions. As shown in Figure 6, the results reflect a high degree of stakeholder satisfaction with the platform's performance and its alignment with digital transformation goals.

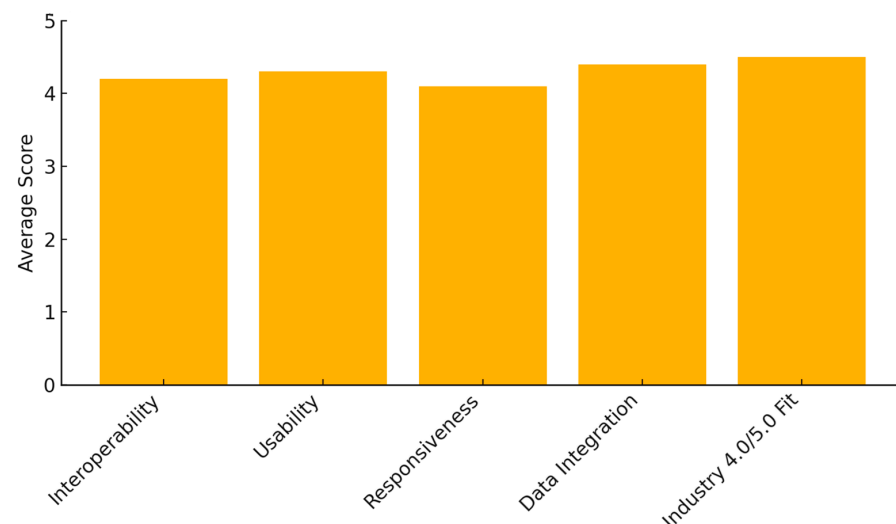


Figure 6. Summary of average scores from the validation questionnaire evaluating platform performance across five key categories.

The proposed framework was implemented in a Portuguese Extended Company that includes three factories that interoperate and collaborate on the overall Company's decision-making process. The implemented framework was validated by using a questionnaire consisting of 10 questions evaluated by 32 participants (including operators, managers, and decision-makers), designed to assess integration, agility, and decision-making efficiency, including not just experts, directors and decision-making staff, but further local Company operators.

The validation involved 32 participants. This group of 32 participants from the implementing Portuguese manufacturing group included operational-level users (e.g., production line operators), middle managers (e.g., plant coordinators, IT staff), and strategic-level decision-makers (e.g., directors and process engineers). Their varied roles ensured that the evaluation captured both usability concerns and strategic performance outcomes.

The questionnaire consisted of 10 items, each of which aligned with a specific technological or organizational aspect of the proposed framework. Sample questions included:

- "To what extent does the platform enable real-time monitoring of key manufacturing indicators?"

- “How effective is the framework in supporting collaborative decision-making across departments or factories?”
- “Does the system improve integration with external stakeholders (e.g., suppliers, partners)?”
- This demonstrates that your questions were well-designed and aligned with framework objectives.

The detailed scores (number of points) about the main framework’s (integrated platform) features assessed the underlying manufacturing management technology and tools used. The corresponding management paradigms, technologies or tools are summarized in Table 5.

Table 5. Summary of the questionnaire results.

Question	Management Paradigm (Technology/Tool)	Number of Points (Assessment Order)
Q1	Dynamic	138 (4)
Q2	Augmented/virtual/mixed reality	125 (9)
Q3	Cloud computing	134 (8)
Q4	Real time based	147 (3)
Q5	Cyber physical systems	125 (9)
Q6	Digital twins	136 (6)
Q7	Distributed/Parallel	152 (2)
Q8	Agile	137 (5)
Q9	Integrated	154 (1)
Q10	Intelligent	135 (7)

Figure 7 shows a bar chart of the distribution of points obtained for each of the 10 main questions in the questionnaire about the evaluation of the importance of the main features underlying the implemented framework at the local Company’s factories, regarding its suitability for enabling an effective and improved operation and management (integrated decision-making) process among the interoperating factories, based on the integrated platform tested.

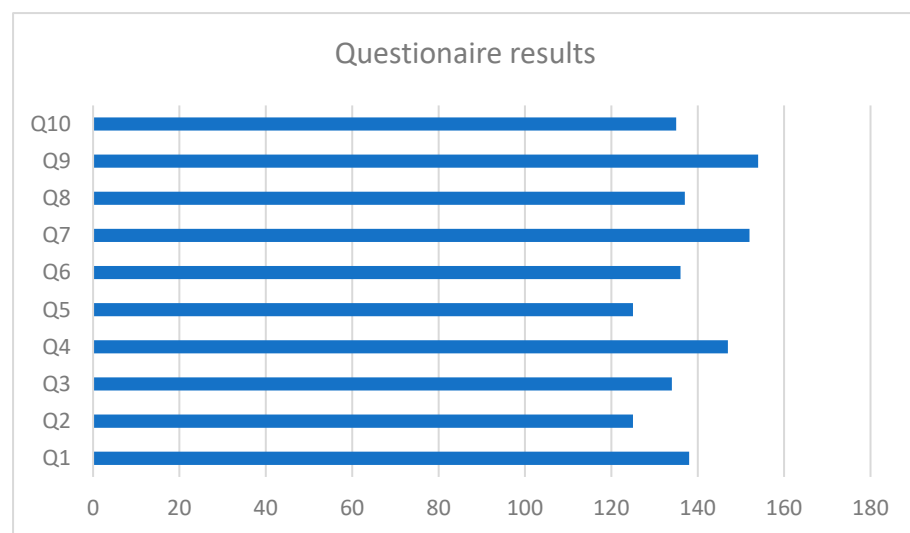


Figure 7. Distribution of participant ratings ($n = 32$) across ten evaluation questions, each mapped to a specific integration paradigm or enabling technology (e.g., real-time control, digital twins, agile management).

These results are further synthesized in Table 6, which outlines the key evaluation dimensions and the assessment methods applied.

Table 6. Summary of Validation Dimensions and Assessment Methods.

Evaluation Dimension	Assessment Method
Real-time integration	Questionnaire (Q4, Q9)
Decision-making support	Questionnaire + Interview
Usability and user satisfaction	Open-ended feedback
External interoperability	Questionnaire (Q6, Q10)
System flexibility and agility	Questionnaire (Q1, Q7, Q8)

According to the results obtained and presented in Figure 4 and Table 4, it can be seen that Q9 (Integrated management paradigm) obtained the highest score (154 points), followed by Q7 (Distributed/parallel paradigm) (152 points), and then Q4 (Real-time based management paradigm) (147 points). Q1 (Dynamic management paradigm) obtained 138 points, followed by Q8 (Agile management paradigm) with 137 points and Q10 (Intelligent management paradigm), with 135 points. Q6 (Digital Twins and Augmented Reality) obtained 136 points, followed by Q10 (Intelligent management paradigm) with 135 points, and Q3 (Cloud computing) with 134 points. Finally, Q2 (Augment/virtual and mixed reality) and Q5 (Cyber-Physical Systems) obtained 125 points.

In addition to quantitative scores, qualitative feedback was gathered through open-ended responses and follow-up interviews. Several participants emphasized the system's flexibility, intuitive dashboard design, and its ability to unify data from previously disconnected systems. However, some concerns were raised about the learning curve for non-technical users and the need for further integration with legacy ERP systems.

Despite these encouraging results, it is acknowledged that the validation is primarily based on user perception, which, although valuable, may not fully capture the platform's objective performance in long-term or high-stress production environments. No A/B testing or real-time benchmarking was conducted in this initial phase. Therefore, future work will involve experimental validations using performance indicators such as decision latency, throughput time, and system availability under live operational conditions.

5. Discussion

As discussed earlier, existing approaches often lack full organizational integration or are limited in scope and validation, supporting the need for the proposed framework that bridges technological advancement and operational impact.

The integration of manufacturing management functions has long been a focal point of industrial evolution. Early efforts emphasized physical integration through methodologies, like just-in-time (JIT) and total productive maintenance (TPM), which aimed to optimize material flow, reduce downtime, and enhance overall operational efficiency. These foundational strategies paved the way for more advanced integration approaches.

The advent of information technology (IT) brought about a paradigm shift, with enterprise resource planning (ERP) systems becoming a cornerstone of information integration. These systems unified core business processes, such as inventory management, procurement, and financial planning, providing manufacturers with centralized data and streamlined operations.

Building on ERP's success, newer technologies and frameworks have further advanced the integration landscape. Manufacturing execution systems (MES), for instance, enable the real-time monitoring, tracking, and control of production activities, effectively bridging the gap between ERP systems and shop floor operations. This ensures that decision-makers have immediate visibility into production performance, enabling timely interventions and optimization.

Additionally, lean manufacturing principles have gained widespread adoption, focusing on the elimination of waste, the optimization of workflows, and fostering a culture of continuous improvement across all organizational functions. Complementing lean, agile manufacturing has emerged as a strategy to enhance flexibility and responsiveness, enabling companies to adapt quickly to shifting market demands, customer preferences, and technological advancements.

Together, these methodologies and technologies have significantly transformed manufacturing management, enabling companies to achieve higher efficiency, resilience, and competitiveness in an increasingly dynamic global market.

Currently, there is a shift in manufacturing management due to the use of advanced manufacturing paradigms and the underlying technologies and tools that promote and foster integrated manufacturing management processes through distributed manufacturing environments or in other, more or less closely related, extended, virtual or collaborative manufacturing contexts.

In this regard, the study conducted and the framework developed and validated, are intended to enable the integration of manufacturing management processes—particularly management paradigms, such as distributed or parallel, dynamic, agile, intelligent, and cloud-based paradigms, along with other associated technologies and tools for enabling real-time based data acquisition, processing and analysis.

Further, the use of digital twins, along with augmented reality, and other Industry 4.0 technologies, permits and fosters collaborative manufacturing management decision-making processes, are crucial to considering and enabling integrated processes in terms of the Company's overall management—from the strategical to the operational manufacturing management level—but also its further integration with other stakeholders and management issues, not just at a local, and internal level, but in an extended and overall collaborative context. For example the integration of approaches for total quality management (TQM) with manufacturing management, has also been increasing over the past decade, not just in academic but also in practical domains.

Thus, the current era of Industry 4.0 brings new dimensions to integration. The convergence of technologies, like IIoT, cloud computing, and AI, enables real-time data exchange and analysis across the entire value chain. This allows for more sophisticated forms of integration such as:

- Vertical integration: Connecting different levels of the manufacturing hierarchy, from shop floor devices to enterprise-level systems;
- Horizontal integration: Linking different companies within the supply chain, enabling collaborative planning and execution;
- End-to-end integration: Connecting all stages of the product lifecycle, from design and engineering to manufacturing and after-sales service.

Challenges to integration remain. These include data security concerns, interoperability issues between different systems and factories, the need for a skilled workforce to manage new technologies, and the organizational changes required to support fully integrated processes.

Although significant achievements have been made in integrating manufacturing management functions, there are many opportunities for further development. The full enablement of the transition to Industry 4.0 requires tackling challenges like data security, interoperability between disparate systems, and fostering a culture of collaboration across all organizational levels. Additionally, research should explore how to leverage further advanced analytics and artificial intelligence approaches and tools to enable additional contributions for optimizing decision-making across the entire manufacturing ecosystem. Furthermore, the human factor in this technological transformation should not be neglected.

Training and upskilling the workforce to effectively utilize these new technologies is crucial for successful integration and realizing the full potential of Industry 4.0 and Industry 5.0, where the former focuses on automation and digitalization and the latter emphasizes human-centric, resilient and sustainable approaches.

In summary, the integration of digital twins, cyber-physical systems, blockchain, AI/machine learning, and other technologies and approaches available through integration platforms creates a robust ecosystem that will drive the evolution of modern manufacturing towards smarter, more agile, and sustainable practices. These technologies work in synergy to enhance real-time decision-making, collaboration, and system-wide optimization in complex, distributed manufacturing environments.

While the implementation of integrated systems presents challenges, including substantial upfront costs and the demand for specialized technical expertise, their long-term benefits are compelling. These systems can streamline operations by automating workflows and improving process efficiency, leading to significant cost reductions in production and maintenance. Additionally, their adaptability enables manufacturers to respond quickly to market changes, incorporate new technologies, and scale operations effectively. As a result, investing in such systems is increasingly essential for manufacturers striving to maintain a competitive edge in a rapidly evolving industrial landscape.

The proposed framework has demonstrated value not only at the operational level but also in supporting broader strategic objectives. Operationally, it streamlined data flows, enabled real-time decision-making, and improved responsiveness to disruptions. Strategically, it allowed for more integrated planning across sites, fostered collaboration between internal and external actors, and supported agility in resource allocation and supply-chain interactions. These capabilities align closely with the core priorities of Industry 4.0 and Industry 5.0—particularly with respect to sustainability, resilience, and human-centricity.

Despite its demonstrated benefits, the framework presents certain limitations. First, it was validated within a single industrial context (a Portuguese multi-factory group), which may limit generalizability to other sectors or organizational models. Second, while the architecture supports modularity, large-scale implementations may encounter integration challenges, especially when interfacing with legacy ERP or MES systems. Additionally, although the user interface is designed for accessibility, a learning curve remains for less technically experienced personnel. Finally, the current validation methodology relies on user perceptions and self-reported gains. Future work will involve objective performance measurements such as benchmarking system throughput, decision latency, and process compliance before and after implementation.

Addressing these limitations will require not only technical refinements but also deeper engagement with change management and cross-industry trials.

6. Conclusions

This study proposes and validates a comprehensive framework for the integration of manufacturing management processes, bridging strategic and operational levels within and beyond the boundaries of a single company. Designed in response to key challenges, such as integration complexity, interoperability, real-time coordination, and data security, the framework supports agile, collaborative, and data-driven decision-making in both local and distributed manufacturing environments.

The solution integrates a diverse set of enabling technologies—including digital twins, cyber-physical systems, AI/machine learning, cloud-edge computing, blockchain, and collaborative platforms—aligning with Industry 4.0 and 5.0 principles. It was implemented and validated in a Portuguese manufacturing group comprising three interoperating factories. The validation results, based on both quantitative and qualitative data collected

from a cross-section of stakeholders, confirm the platform's effectiveness in enhancing decision-making, stakeholder collaboration, and operational responsiveness.

Strategically, the framework fosters alignment between digital transformation initiatives and organizational goals, supporting sustainable, human-centric, and resilient manufacturing practices. Operationally, it streamlines communication, integrates diverse technologies, and enables responsiveness to market fluctuations and process deviations.

Nevertheless, limitations remain. The current validation is perception-based and context-specific; further empirical studies are required to benchmark performance across different industrial sectors. Future work will focus on scaling the framework to other domains, refining AI-driven decision mechanisms, and incorporating objective performance metrics such as latency, resource utilization, and return on investment (ROI).

The structured survey provided valuable feedback from stakeholders, confirming that the proposed solution meets essential criteria for integration, responsiveness, and alignment with Industry 4.0/5.0 paradigms.

In conclusion, the proposed framework represents a robust step toward integrated, intelligent, and sustainable manufacturing management, helping companies to navigate the demands of digital transformation and competitive global markets.

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